PROBLEM–SOLUTION FIT

ComicCraft – AI Comic Story Creator using Gemini Models

TEAM INFORMATION	
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Team Size	4
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PROJECT OVERVIEW

ComicCraft is a web-based AI application that generates personalized comic stories and illustrations from user prompts. The application uses Gemini models for story outlines, narration and dialogue, and Stable Diffusion for comic-style illustrations. Users can preview the generated comic and download the complete comic as a PDF.

Core technologies: FastAPI, HTML/CSS/Jinja2, Google Gemini models, Hugging Face Diffusers/Stable Diffusion, PyTorch and FPDF.

OBJECTIVE

To make comic creation accessible to non-artists, storytellers and hobbyists by automating story outlining, narration, dialogue, illustration generation and PDF export.

PROBLEM–SOLUTION FIT CANVAS

1. CUSTOMER SEGMENT(S) [CS]

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- Non-artists and casual creators
- Storytellers and hobbyists
- Users who want personalized comics without advanced design skills
- Users seeking quick AI-assisted visual storytelling

2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

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- Creating a complete comic manually requires significant time and creative effort.
- Users may lack illustration or drawing skills.
- Turning an idea into a coherent panel-by-panel story can be difficult.
- Generating matching visuals, narration and dialogue separately can be inconvenient.
- Frequency: The problem can occur whenever a user wants to create a new comic story.

3. TRIGGERS TO ACT [TR]

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- A user has a comic idea but needs help turning it into a complete story.
- The user wants personalized characters, settings, tone and art style.
- The user wants to quickly change the mood or visual style and regenerate the comic.
- The user wants to save, print or share the finished comic as a PDF.

PROBLEM–SOLUTION FIT CANVAS – CONTINUED

4. EMOTIONS BEFORE / AFTER [EM]

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Before: Users may find it difficult to convert an idea into a coherent comic, especially without art or storytelling expertise.

After: Users can review a personalized panel-by-panel comic with generated narration, dialogue and illustrations, and export it as a PDF.

5. AVAILABLE SOLUTIONS PROS & CONS [AS]

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- Manual comic creation: High creative control, but time- and skill-intensive.
- Separate writing/image tools: Flexible, but requires moving between tools and aligning outputs.
- AI-assisted tools: Faster generation, but quality and consistency can vary.
- ComicCraft: Integrates story, image generation, preview and PDF export into one workflow.

6. CUSTOMER LIMITATIONS [CL]

6. CUSTOMER LIMITATIONS [CL]

- May not have professional drawing or illustration skills.
- May have limited time for manual comic production.
- May prefer a simple web interface rather than complex creative software.
- Output quality depends on AI-generated content and model performance.
- Current version focuses on single-comic generation without user account management.

PROBLEM–SOLUTION FIT CANVAS – CONTINUED

7. BEHAVIOR + ITS INTENSITY [BE]

7. BEHAVIOR + ITS INTENSITY [BE]

- Users provide a story prompt and customization details.
- Users experiment with tone and art style.
- Users preview the generated panels.
- Users may regenerate content to better match their creative vision.
- Users download the final comic as PDF.
- Intensity: Higher when users need a complete, personalized comic quickly.

8. CHANNELS OF BEHAVIOR [CH]

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ONLINE:
- Web interface / browser
- FastAPI-powered application
- API endpoints for comic generation
- Local web access during development/testing

OFFLINE:
- Downloaded PDF can be saved, printed and viewed offline.

9. PROBLEM ROOT / CAUSE [RC]

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- Comic creation traditionally combines multiple creative tasks.
- Story structure, dialogue and visual design require different skills.
- Manual panel creation is time-consuming.
- Users may struggle to maintain consistency between story text and visuals.
- Lack of an integrated workflow increases effort.

10. YOUR SOLUTION [SL]

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ComicCraft provides an end-to-end AI comic creation workflow. A user enters a story prompt, character name, setting, tone and art style. Gemini Flash generates a structured 5-panel outline; Gemini Pro expands it into narration and dialogue; Stable Diffusion generates comic-style illustrations; the application assembles the panels into a layout and exports the finished comic as a downloadable PDF. The web interface provides preview and export functionality, making comic creation more accessible to users without professional art skills.

SOLUTION WORKFLOW

Step	Activity	Output / Benefit
1	User enters story prompt, character name, setting, tone and art style.	Personalized comic requirements
2	Gemini Flash creates a structured 5-panel outline.	Panel-by-panel story structure
3	Gemini Pro creates narration and dialogue.	Detailed comic story
4	Stable Diffusion generates illustrations.	Comic-style panel images
5	Layout builder organizes text and images.	Complete comic layout
6	FPDF exports the final comic.	Downloadable multi-page PDF

CONCLUSION

ComicCraft demonstrates an end-to-end approach to AI-assisted comic creation. Its integrated workflow addresses the effort involved in story planning, dialogue writing, illustration and final document creation. The project document also identifies future enhancements such as user profiles, comic libraries and multi-page story arcs.

Source basis: Prepared from the supplied ComicCraft project document and the supplied Problem-Solution Fit canvas template.